

Tenacity Tutoring

Quadratic Equations

Year 10 maths

Name: Sample Student Date: _____

Learning Objectives

- Solve quadratics by factorising.
- Apply the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Solving by factorising

If a product $ab = 0$ then $a = 0$ or $b = 0$. Rewrite $x^2 - 5x + 6$ as $(x - 2)(x - 3)$.

Definitions

Term	Definition
Root	A value of x for which the expression equals 0.

Worked Examples

Example 1: $x^2 - 5x + 6 = 0$

Working	Explanation
$x^2 - 5x + 6 = 0$	Start with the equation.
$(x - 2)(x - 3) = 0$	Factorise: two numbers multiplying to 6, adding to -5 .
$x = 2$ or $x = 3$	Apply the null factor law.

Example 2: the quadratic formula

Working	Explanation
$2x^2 + x - 23 = 0$	Here $a = 2$, $b = 1$, $c = -23$.
$x = \frac{-1 \pm \sqrt{1^2 - 4(2)(-23)}}{2(2)}$	Substitute into the formula.
$x = \frac{-1 \pm \sqrt{185}}{4}$	Simplify the discriminant.
$x \approx 2.65$ or $x \approx -3.15$	Evaluate to 2 d.p.

Tip: Always check both roots by substituting back.

Common Mistake: Forgetting the $\pm$ gives only one root.

Practice Questions

1. Solve $x^2 - 5x + 6 = 0$ by factorising.

3. Simplify $\frac{3}{4} + \frac{2}{5}$ and express $\sqrt{50}$ in simplest surd form.

End of Topic Quiz

Quiz

4. Solve $x^2 = 49$.

.....

5. Simplify $\sqrt{72} - \sqrt{18}$.

.....

Answers

Solving by factorising

Q#	Answer
Q1	$x = 2$ or $x = 3$
Q3	$\frac{23}{20}; 5\sqrt{2}$

Quiz

Q#	Answer
Q4	$x = \pm 7$
Q5	$3\sqrt{2}$